



AI Powered Analytics using Power BI MCP Server and Claude

HOUSTON AREA SQL SERVER USER GROUP | SEPTEMBER 2, 2026

Sunil Kadimdiwan

About me: Sunil Kadimdiwan

Background

- Focus on Data Analytics for 25+ years
- Microsoft AI/BI stack, Snowflake, Databricks

Community Involvement

- Frequent speaker at Data Saturdays and User Groups

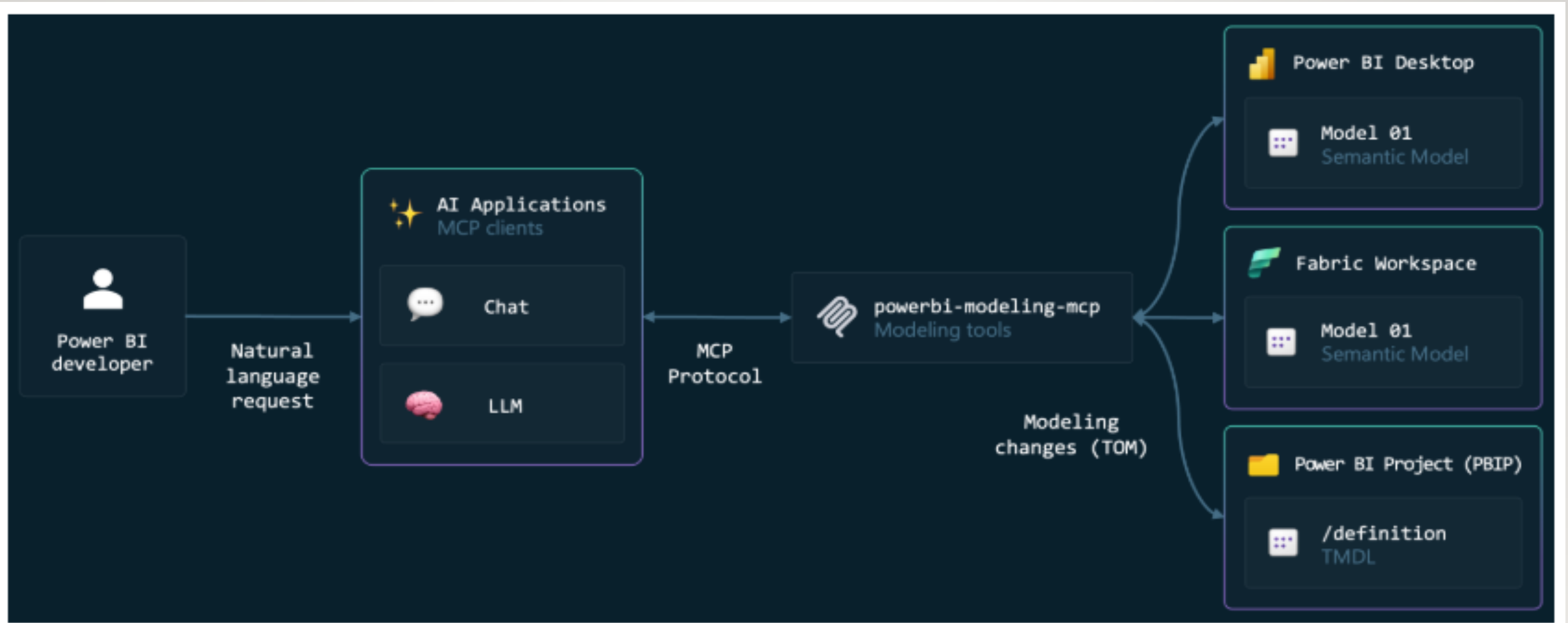
Contact info



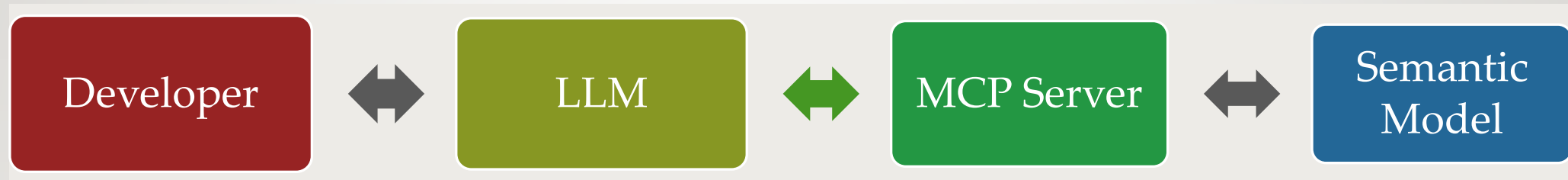
@sunilkadimdiwan

Agenda

- What is Power BI MCP (Model Context Protocol) Server?
- Demos
- Useful Links



Overview: Microsoft Power BI MCP Server



WHAT CAN WE DO?

Natural Language Editing

Bulk Operations

DAX Query

Documentation

KEEP in MIND

Data and model sent to LLM

AI Hallucinations, verify results

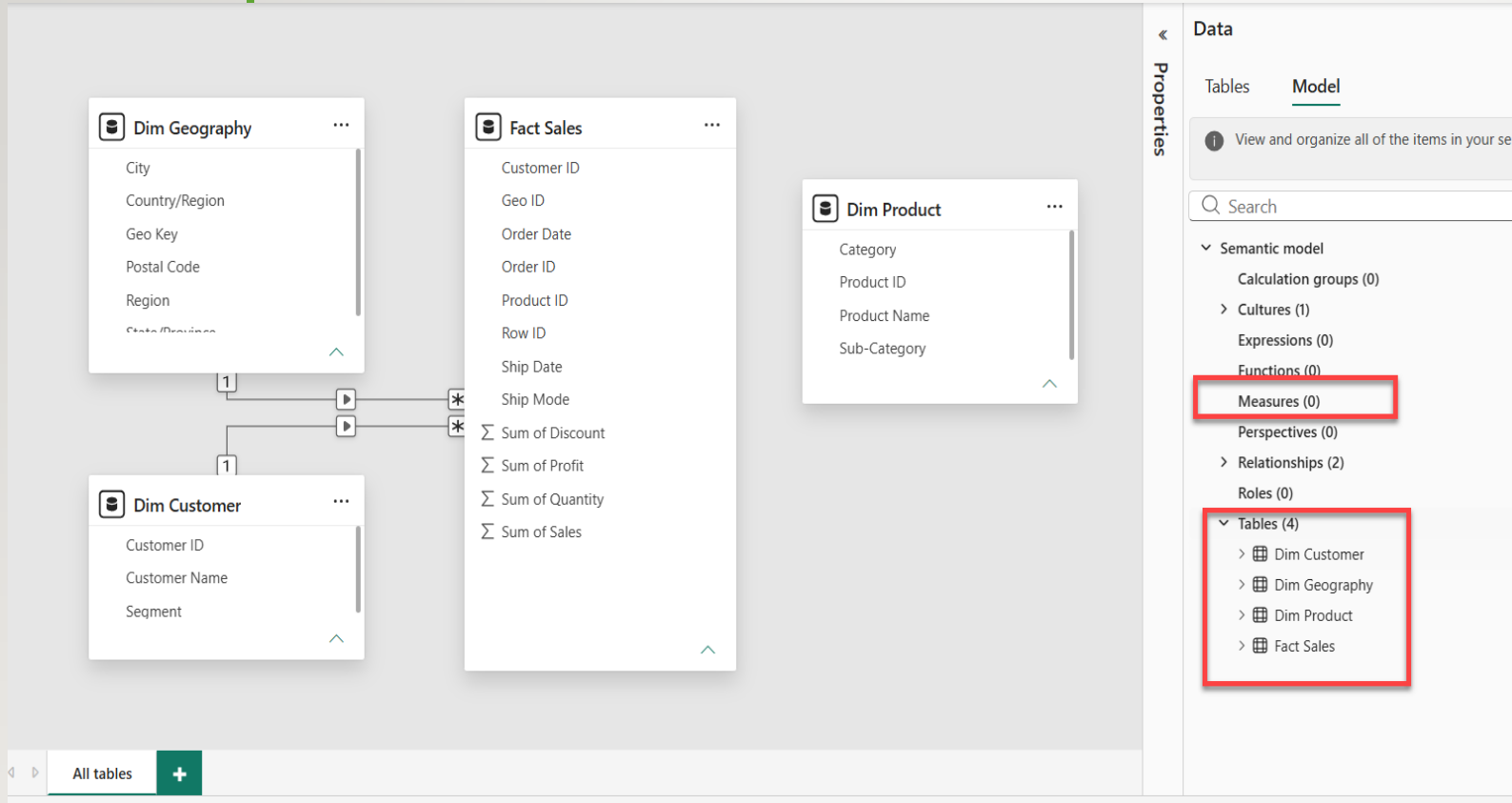
Have appropriate permissions

MCP Server In Public Preview

Review Semantic Models

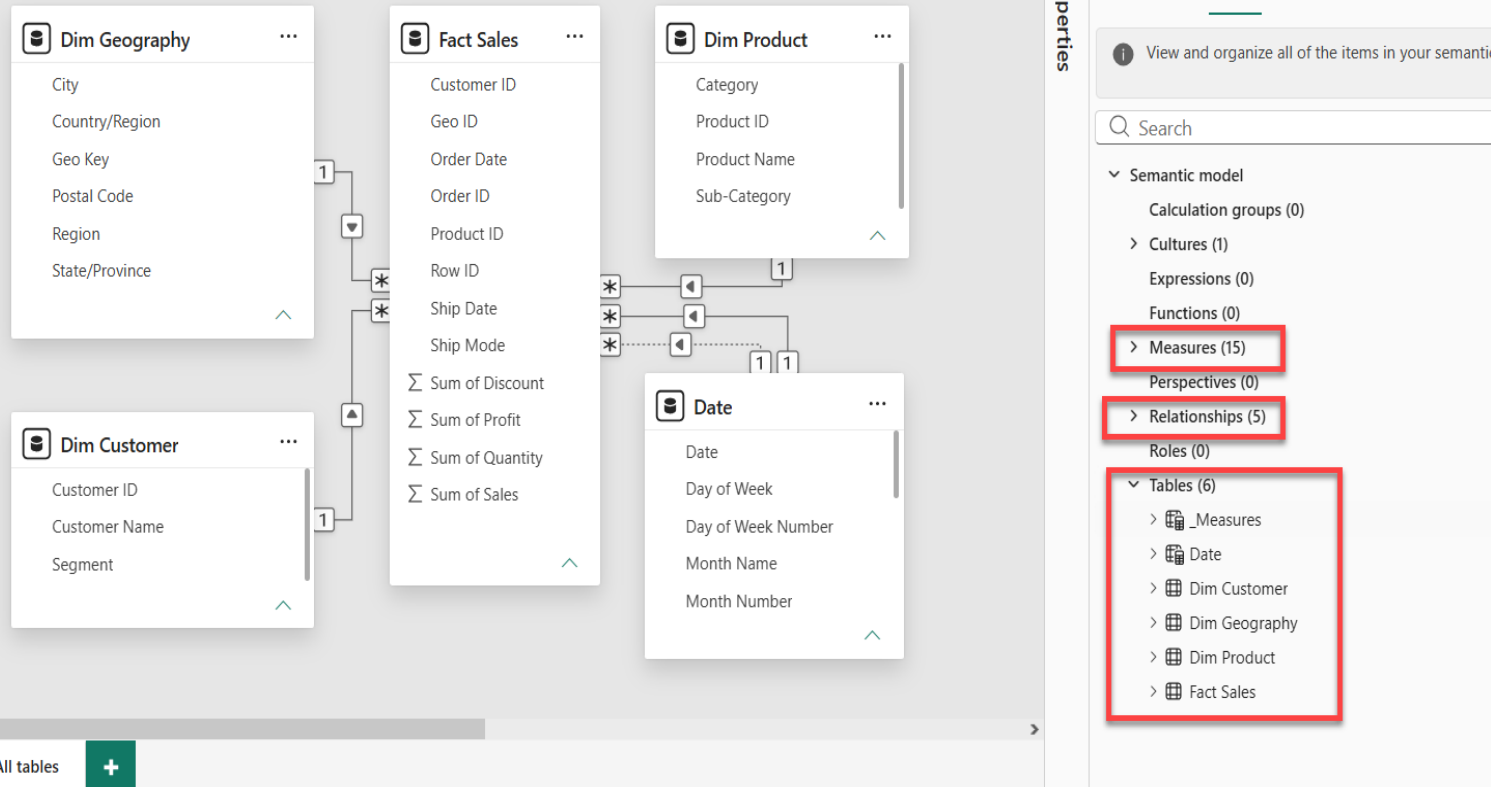
Initial and Enhanced

Initial Data Model



- Missing Product relationship
- Missing Date Dimension
- ZERO Measures
- Missing Documentation

Enhanced Data Model



- Establish Product relationship
- Create Date Dimension
- Sum Measures
- Date/Time Measures
- Document Measures
- Add Documentation

Demo how to **Enhance** the Semantic Model

Using Prompts for Claude

How can YOU try this?

List out the steps

Try this on your laptop?

- Install Power BI Desktop
- Install VS Code
 - Install Extension: Power BI MCP Server
- Install Claude Desktop
 - Configure to use Power BI MCP Server
 - Maybe free account is okay, I have paid \$20 / month account
- Sample data, I got one from
 - <https://www.kaggle.com/datasets>



Link to GitHub Repo:
<https://tinyurl.com/mvz7khca>

Walk Thru

- Installed components on laptop
- GitHub Repo files

Links for learning

- <https://github.com/microsoft/powerbi-modeling-mcp>
- I learn a lot from YouTube videos, thanks to these folks
 - <https://www.youtube.com/watch?v=jDSoSJz4ams&t=321s> (Power BI Bro)
 - <https://www.youtube.com/watch?v=WPVitWrLk-Y> (How to Power BI)
 - <https://www.youtube.com/watch?v=YdymjIHymdQ> (Pragmatic Works)